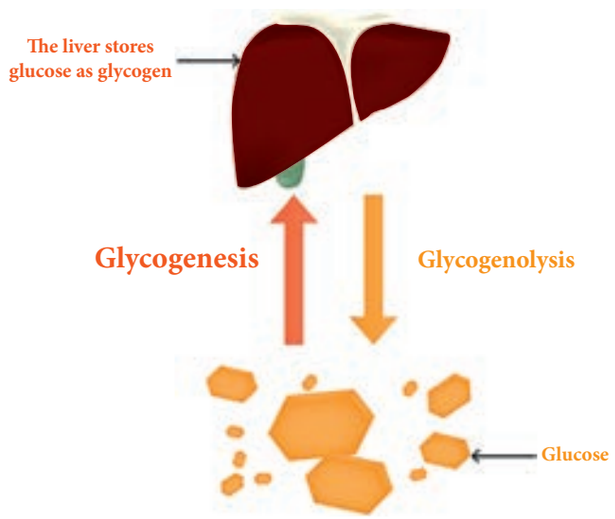
A third enzyme, glycogen debranching enzyme transfers three of the remaining four glucose units to the end of another glycogen branch, exposing the α [1→6] branching point. This glycosidic bond is hydrolysed by a fourth enzyme α [1-6] glucosidase which eliminates the branch by removing the final glucose as a molecule of glucose, rather than glucose-1-phosphate. In muscle, but not liver cells, the glucose is subsequently phosphorylated to glucose-6-phosphate by a fifth enzyme hexokinase and enters the glycolytic pathway.

In liver cells, the main purpose of the breakdown of glycogen is for the release of glucose into the bloodstream for uptake by other cells. The phosphate group of glucose-6-phosphate is removed by the enzyme glucose-6-phosphatase and the free glucose exits the cell via the membrane localized GLUT2 glucose transporter.

In muscle, glycogenolysis serves to provide an immediate source of glucose-6-phosphate for glycolysis to provide energy for muscle contraction but not for other body tissues. Muscle cells lack the enzyme glucose-6-phosphatase and thus cannot convert glucose-6-phosphate (which cannot be transported across the cell membrane) to glucose.



3.7. GLUCONEOGENESIS

Gluconeogenesis is a process through which noncarbohydrate precursors such as lactate, pyruvate, glycerol, and amino acids are converted to glucose.

Glucose occupies a central role in metabolism, both as a fuel and as a precursor of essential structural carbohydrates and other biomolecules. The brain and red blood cells are almost completely dependent on glucose as an energy source. Yet the liver's capacity to store glycogen is only sufficient to supply the brain with glucose for about half a day under fasting or starvation conditions. Thus, when fasting, most of the body's glucose needs must be met by gluconeogenesis (literally, new glucose synthesis), the biosynthesis of glucose from noncarbohydrate precursors.

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